

Miscellaneous/Review

Body

1. (August 2020 Problem 1) On this problem, no justification is required; only the answer will be graded.

- (a) Give an example of a non-diagonalizable matrix in $\mathrm{GL}_3(\mathbb{F}_p)$ (for some p).
- (b) Give an example of a nontrivial proper normal subgroup of $\mathrm{GL}_3(\mathbb{F}_p)$ (for some p).
- (c) Give an example of a one-sided ideal (either left or right) in $M_2(\mathbb{C})$ which is not a two-sided ideal.
- (d) Give an example of a $\mathbb{Z}$ -module which is flat but not free.
- (e) Give an example of an exact sequence of abelian groups which is not split.

2. (January 2021 Problem 1) On this problem, no justification is required; only the answer will be graded.

- (a) Give an example of a finite extension of $\mathbb{Q}$ which is not Galois.
- (b) Give an example of a non-trivial, proper, two-sided ideal in $M_2(\mathbb{Z})$.
- (c) Give an example of an injective map $M \rightarrow N$ of $k[x]$ -modules (k a field) and another $k[x]$ -module R such that $M \otimes R \rightarrow N \otimes R$ is not injective.
- (d) Give an example of a commutative ring R and prime ideals p, q, q' in R such that the extension $\bar{q}$ of q to the localization R_p is prime, while the extension $\bar{q'}$ of q' to R_p is not prime.

3. (January 2020 Problem 1) On this problem, no justification is required; only the answer will be graded. Let S_3 be the symmetric group on 3 letters, and let R be the group ring $\mathbb{Z}[S_3]$.

- (a) Write down a nonzero element of R which is a zero-divisor.
- (b) Write down an element in the center of R which is not in $\mathbb{Z}$.
- (c) Write down an R -module which is not a free module.

4. (January 2017 Problem 3) Let R be a commutative ring with unity. Show that a polynomial

$$f(t) = c_n t^n + c_{n-1} t^{n-1} + \cdots + c_0 \in R[t]$$

is nilpotent if and only if all of its coefficients $c_0, \dots, c_n \in R$ are nilpotent.

Soul

5. (January 2021 Problem 4) Let $F/\mathbb{Q}$ be a quadratic extension, and let f and g be two maps of $\mathbb{Q}$ -algebras from F to $M_2(\mathbb{Q})$. (Edit: in this problem we take algebra homomorphisms to be *unital*, meaning that $f(1) = g(1) = I$.)

- (a) Prove that there exists a matrix $X \in \mathrm{GL}_2(\mathbb{Q})$ such that for all $\alpha \in F$, $Xf(\alpha)X^{-1} = g(\alpha)$.
- (b) Prove that there does not exist a choice of f whose image is contained in the set of upper triangular matrices.

6. (January 2018 Problem 3) Consider the field with 11 elements $F = \mathbb{F}_{11}$. Let G denote the cyclic group of order 11 (written multiplicatively), with generator $r \in G$. Denote by FG the group algebra of G (also sometimes denoted $F[G]$). We consider r as an element of FG , and let $T: FG \rightarrow FG$ be the F -linear map given by $T(x) = rx$ for all $x \in FG$. Find the Jordan canonical form of T .

7. (January 2023 Problem 4) Consider the finite field $\mathbb{F}_p$ where p is a prime number.

- (a) How many monic irreducible polynomials of degree 5 are there over $\mathbb{F}_p$? (Your answer should be a function of p .)
- (b) Let $P(x), Q(x) \in \mathbb{F}_p[x]$ be irreducible polynomials. Give a necessary and sufficient condition for there to exist a third polynomial $R(x) \in \mathbb{F}_p[x]$ such that $Q(x)$ divides $P(R(x))$. (The polynomial $R(x)$ need not be irreducible.)

8. (January 2020 Problem 2) Let k be a field, and let $R = M_2(k)$, the (noncommutative) ring of 2×2 matrices over k .

- (a) Give an example of a simple left R -module M , and a simple right R -module N .
- (b) Can you find another simple left R -module M' which is not isomorphic to M ? (Either find one, or explain why there is none.)
- (c) Compute $\dim_k N \otimes_R M$.

9. (August 2016 Problem 2) Let V be a non-zero finite dimensional vector space over $\mathbb{C}$. Given a linear transformation $A: V \rightarrow V$, show that the following are equivalent.

- (i) There exists a linear transformation $P: V \rightarrow V$ such that $P^2 = I$ and $AP = -PA$.
- (ii) There exists an invertible linear transformation $P: V \rightarrow V$ such that $AP = -PA$.
- (iii) There exists a direct sum decomposition $V = V_1 \oplus V_2$ such that $AV_1 \subset V_2$ and $AV_2 \subset V_1$. (Hint: Thinking about (iii) $\Rightarrow$ (i) helped me to think about (ii) $\Rightarrow$ (iii).)